Module 2 Lecture - Research Sources

Research Methods in Psychology

Quinton Quagliano, M.S., C.S.P

Department of Psychology

Table of Contents

1 Overview **2**
1.1 Overview 2

2 Research vs. Personal Experience **2**
2.1 Overview 2
2.2 No Comparison 3
2.3 Experience is Confounding 4
2.4 Research > Experience 4
2.5 Probability in Research 5

3 Research vs. Intuition **6**
3.1 Bias in Intuition 6
3.1.1 Convincing Narratives 6
3.1.2 Availability Heuristic 6
3.1.3 Present/Present Bias 7
3.1.4 Confirmation Bias 7
3.1.5 Bias Blind Spot 7
3.2 Intuitive Thinking vs Scientific (Empirical) Thinking 8

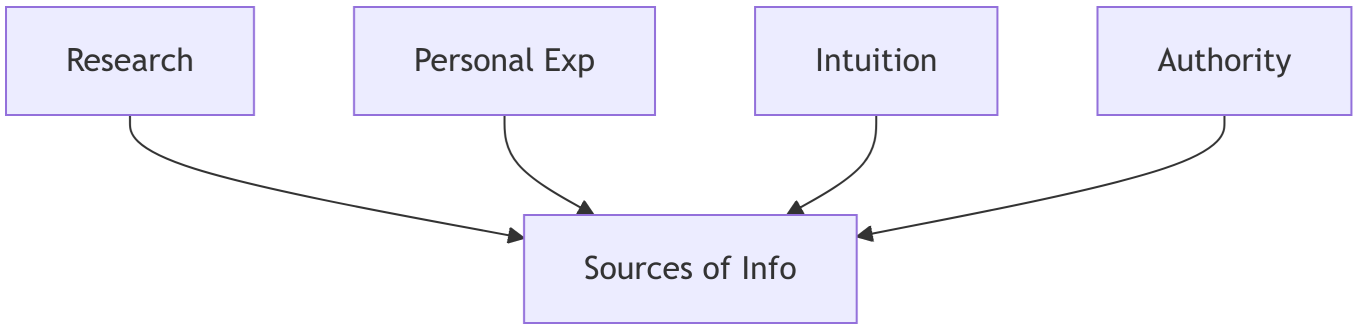
4 Research vs. Authority **8**
4.1 Faults in Appealing to Authority 8
4.2 Flow Chart of Claims 9

5 Examining Research Sources **9**
5.1 Types of Articles 9
5.2 Types Books 10
5.3 Finding & Reading Research 10
5.4 Scientific Journalism's Role 10

1 Overview

1.1 Overview

- We can gain _____ from many origins; this is true of both scientific and common sources and:
  - Scientific Sources:
    - Journal _____
    - Books and edited books
    - Scientific _____
    - etc.
  - Common sources:
    - _____ outlets (and lots of them!)
    - Newspapers
    - Think tanks (private "research")
    - "Well my friend said…"
- *But*, how do we _____ these sources - which ones are the "best" for drawing conclusions? (hint: research!)
  - Also, how do we read each of these _____ more critically?
- We'll compare 4 categories of sources:



2 Research vs. Personal Experience

2.1 Overview

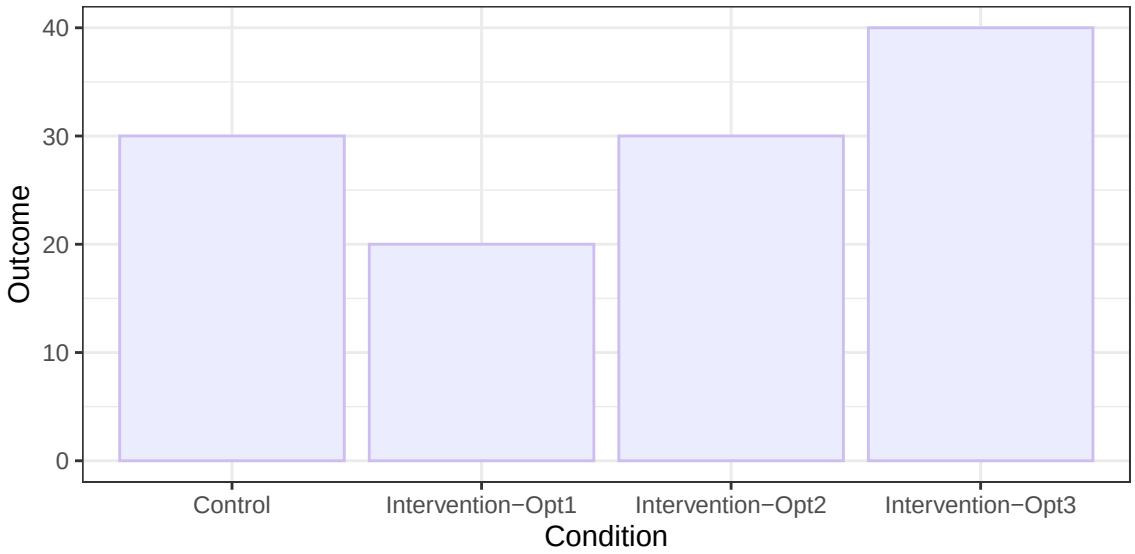
- Personal experience is simply the _____ that comes at us from everyday life.
- This is when we learn from the _____ that occur to and around us.
- *Example:* I learn to walk carefully (like a penguin!) on ice because I have

slipped in the past when walking normally. But does walking like a penguin actually work for people to avoid slipping?

- We often rely on personal experience to make _____, but this does not always mean it is the most reliable source of information…

2.2 No Comparison

- A **comparison group** is a key _____ between scientific research and personal experience. We must compare our study results to some other benchmark, whether that is a _____ group (no intervention) or another type of intervention.
- Without a comparison group we have three _____ outcomes:
  - Our effect is _____ than control (no intervention) or other interventions
  - Our effect is _____ to control or other interventions
  - Our effect is _____ than control or other interventions



- **Example:** A researcher performs a social intervention to reduce feeling of hostility towards an "out-group". It seems like the participants don't harbor any major negative feelings towards the out-group at the end of the study. A success!? Maybe…
  - Why are the _____ (potentially) wrong here? What other comparisons are possible?
- You may have some belief about the _____ of an intervention (from personal experience), but you won't have good _____ of

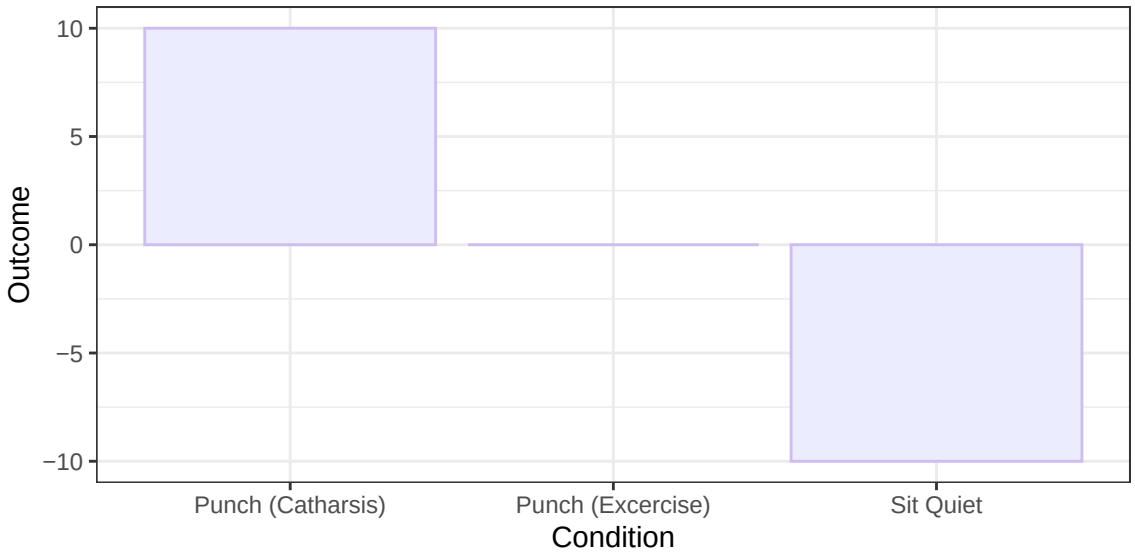
its “real” effect, without a comparison group.

2.3 Experience is Confounding

- Why did something happen? _____ make it difficult to determine causation in our personal experiences. Confounds are variables that may change how or why something happens.
- Example: A professor addresses you in a disappointed tone on the first day of class. You assume it is due to them just disliking students.
  - Why are the _____ (potentially) wrong here? What other explanations are possible?
- Because of confounds, there may be these _____:
  - The professor had a rough day at home and it taking it out on students (unfairly)
  - You forgot your first assignment, and were late
  - The professor has mistaken you for a different student
  - Any number of other possibilities!
- In our own lives it is difficult to impossible to isolate individual causes for certain outcomes, but in research, we have _____ and statistical controls for confounds

2.4 Research > Experience

- In research settings, we can better control for the _____ of confounds, and create comparison groups
- **Example:** Bushman’s study (catharsis theory)



- Another tale of why scientific process > _____

2.5 Probability in Research

- Individual differences are _____ in theories! There are exceptions/edge cases to all hypotheses and, as we mentioned in the last lecture, we cannot account for all possible scenarios with a single _____.
- Research is always _____ which means it try to capture the majority of experiences or the net average of all cases
  - Which may mean your experiences may contradict the _____ of evidence in the research body - but that doesn't mean that your experience or the research consensus is "wrong"
- Connect to your previous statistics class: conclusions are based on p-values, which, in turn, are essentially a _____ of a certain outcome.
- A single case does not fully invalidate a theory or hypothesis. Instead, it may represent a unique case that differs in some way than the others. We must research _____ to understand why!
- *Example:* A medical doctor understands that a blood pressure drug works for _____ individuals with hypertension - but one of their patients isn't responding to it.

3 Research vs. Intuition

3.1 Bias in Intuition

- Recall that _____ is often like a "hunch" or a "gut feeling". We may describe intuition as being "pre-scientific."
- Unfortunately, our intuition may be swayed by certain sources of _____ in how we think…

3.1.1 Convincing Narratives

- Have you ever met someone who could really _____ a story? Maybe a friend who is really good at making stuff up or a salesperson.
- Freud certainly could! Many of his early theories drew on observations that *seemed* reasonable _____, and his psychoanalytic theory is prevalent to this day.
  - This is a sort of "just-makes-sense" thought process, where we think - "well, naturally, it just _____ that way"
- However, this intuition can be an incorrect _____ and may prevent us from fully exploring whether things are actually as we think they are
- *Example:* "Scared straight" program for scaring kids out of certain lifestyles

3.1.2 Availability Heuristic

- *Example:* Anyone afraid of shark attacks in the ocean?
- You may think that they happen often, but really, they are exceedingly _____. But because of popular media and news hype, the idea of shark attacks is much more readily **available** cognitively.
- A **heuristic** is just a cognitive _____, it allows us to skip the process of thinking to come to conclusion. However, in the scientific theory-data-hypothesis cycle, it threatens to obscure what is actually going on
- This heuristic often causes over- or under- estimations of certain situations occurring

3.1.3 Present/Present Bias

- The **present/present bias** may be best understood as an extension of failing to account for _____ groups (see No Comparison)
  - If we don't see a possible outcome, do we fully account for it?
- *Example:* Remember Harlow's monkeys' third option…
- *Additional Example:* I ran in sneakers A and my run was great because of them! But what about sneakers B?
- This may also be understood as also reflecting a _____ to recall certain (critical) events, but not others

3.1.4 Confirmation Bias

- This is our tendency to _____ our already closely-held beliefs by selectively praising, criticizing, and noticing information that reinforces the belief
  - *Especially* relevant to situations like politics, where people have a great motivation to protect "their" _____
- This often causes _____ bias in evidence-gathering and is likely to lead to a single conclusion: "I'm right"
- *Example:*
  - Google search: "Evidence that the earth is flat" vs.
  - "Evidence of the earth's shape"

3.1.5 Bias Blind Spot

- Recall those last few examples of bias? _____ are just as susceptible to those same fallacies as anyone else.
- It pays dividends to be _____ and curious towards possibilities, even when our intuition, experiences, and even research may say otherwise.
- Especially as scientists, there is a _____ to believe *we* are uniquely resistant to bias, *we aren't*
  - Want to eliminate bias? Prove it with sound science, methods, and reporting

3.2 Intuitive Thinking vs Scientific (Empirical) Thinking

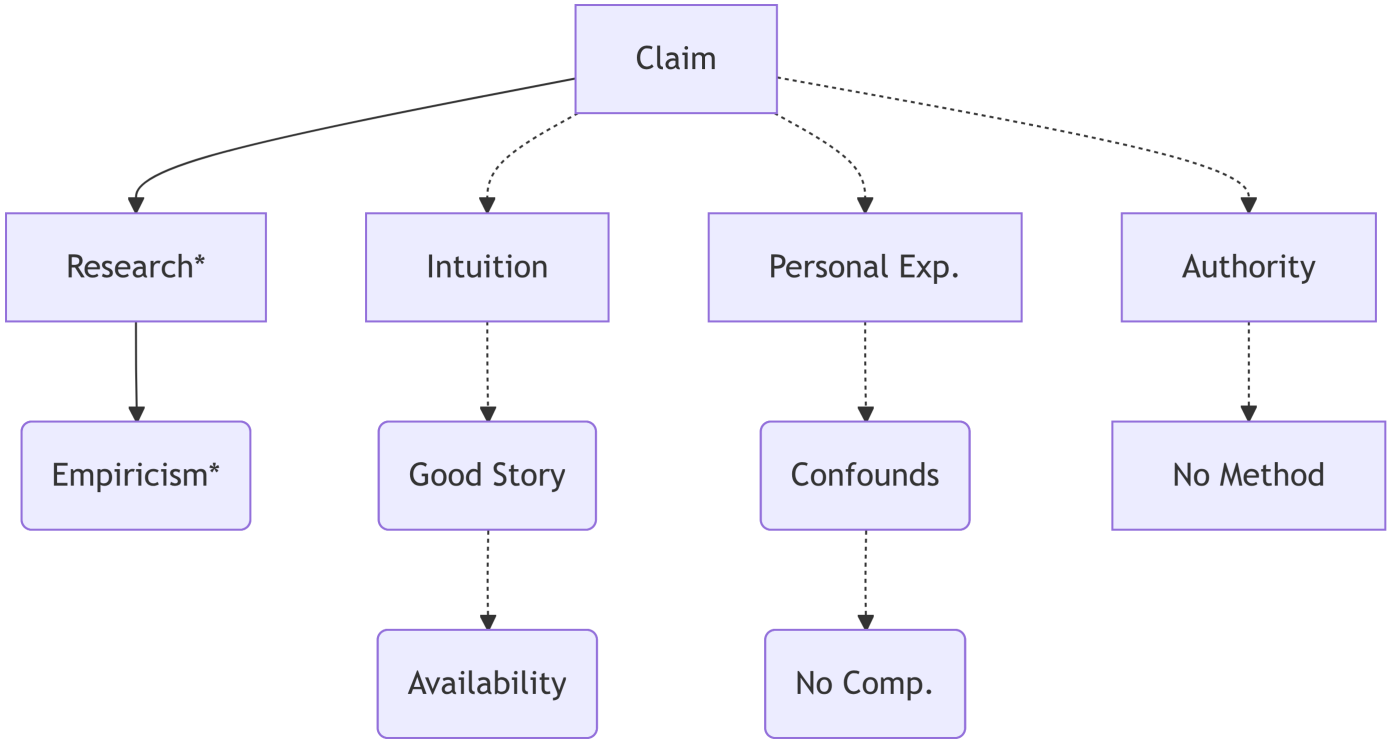
- Taken together, there are lots of ways that intuition and _____ experience can let us down in the general conclusions we _____
- In science, we are interested in using *empirical* principals to guide our thinking and writing, as to capture the overall _____ in data and phenomena (remember, science is probabilistic)
- Simply "conducting research" does not _____ protect us from biases - we must be keenly mindful of the risks on biases in our research processes, starting with literature reviews

4 Research vs. Authority

4.1 Faults in Appealing to Authority

- We tend to _____ many authorities throughout our lives: parents, teachers, professors, textbooks (via their authors), administrators, etc.
- However, we ought to distinguish between trusting a _____ for their perceived expertise and trusting *a process* for producing sound claims
  - Why do we trust journal articles? It isn't just because the researchers are simply _____, but rather, that the peer-review process is well set up to prevent poor science from slipping through
- Therefore, we must _____ *the method* by which a person comes to a conclusion, rather than basing our opinion of the person. This is the purpose of written articles, to interrogate the *how*, rather than the *who*

4.2 Flow Chart of Claims



5 Examining Research Sources

5.1 Types of Articles

- **Original Empirical Journal Articles:** Some form of _____ study in which a scientific process and analysis were conducted. These are often some type of novel observation or _____. Usually contains some form of introduction, methods, results, and discussion.
- **Literature Review Journal Articles:** A comprehensive literature review that _____, synthesizes, and compounds the many available empirical studies in a specific research area. Often, comes across as an extended introduction section.
  - Also, a great starting point when doing a literature review for your own _____!
- **Meta-analysis:** An extension of a literature review articles that calculates a _____ effect size from the surveyed research studies. A more quantitative version of a classic literature review.

5.2 Types Books

- **Scientific Books:** a full-length scientific book written by the same single or multiple authors throughout. Focuses on a single _____, and functions as a sort of extended literature review.
- **Edited Books:** a collaborative effort in which many experts compose _____ on specific topics. These are usually not peer-reviewed to the same extent as journal articles - but the authors are usually some of the most respected researchers in an area. Be careful of appeals to _____ because of this!
  - These are popular among applied practitioners of psychology (_____), because it allows them to keep up with developments and ideas in a nice range of areas.

5.3 Finding & Reading Research

- We will cover this in a separate workshop/presentation! This is a critical skill to develop as an early researcher, and is fairly complicated.
- We will also cover how to determine the sections of articles, and what questions you should ask of yourself in each section.

5.4 Scientific Journalism's Role

- Empirical articles will *always* be the _____ way to gain scientific knowledge. Edited books, literature reviews, meta-analyses, etc. may all be good _____ sources as well.
- However, popular scientific journalism may help you find new, _____ findings to your interests - just don't cite it directly. Instead, go to the original articles.

The instructor-provided glossary may not include all terms worth memorizing, make sure you consider using the vocabulary list in you book and your own judgment to make sure you have all relevant terms